port and controller for easily communicating with your computer. This USB connectivity and friendly sockets for hookup wires to easily plug in to, contribute to the high level of usability in the interface board design.

Equally important to this ecosystem is the Arduino development environment, a program based on the Processing development environment (`http://processing.org`) that you use to write, edit, compile, and upload your Arduino source code to the interface board. The Arduino team made the general assumption that people don't really care about the myriad of technical specifics involved with microcontroller architecture—they just want it to do something cool. With that in mind, the Arduino development environment wraps up some of the more techie parts of programming AVR microcontrollers into a nice, simple library of Arduino-specific commands that are easier to use and easier to understand, and built right into every sketch written for the Arduino. This development environment is so versatile, that an Arduino interface board is not even needed to use it. Instead, we can use the very same AVR microcontroller as is built onto the interface board, but in an entirely different device—such as the example shown in Figure 1-3 of a microcontroller on a breadboard that has been programmed with an Arduino sketch.

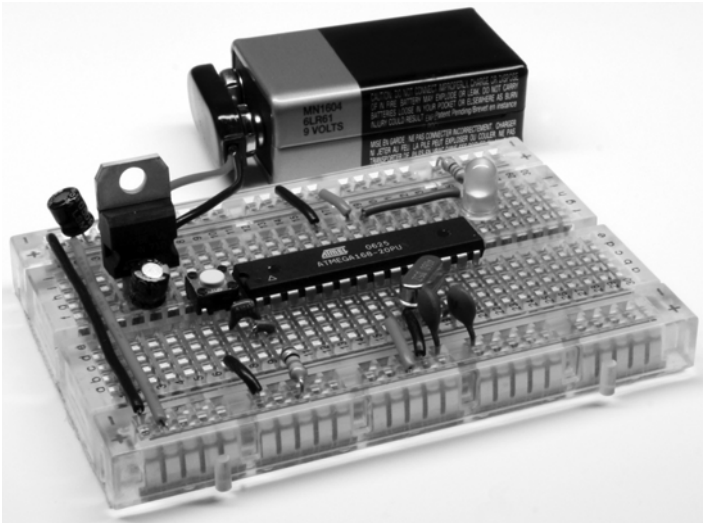


Figure 1-3. *Arduino compatible AVR microcontroller on breadboard*

Finally, we are brought to the Arduino bootloader, a little chunk of code residing on the microcontroller that allows it to be easily programmed with a simple serial connection rather than cranky, external hardware. As long as the bootloader has been programmed onto the microcontroller beforehand, we can use that chip in whatever device we choose, such as the breadboard in Figure 1-3, and still write code for it using the rest of the Arduino platform. It helps if we use one of the microcontrollers common to the Arduino, but this is not always a prerequisite, with many other Atmel microcontrollers working nearly as well with little or no modification.